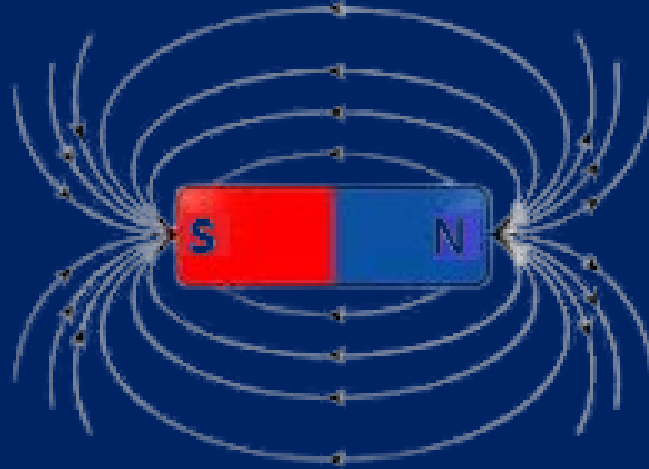


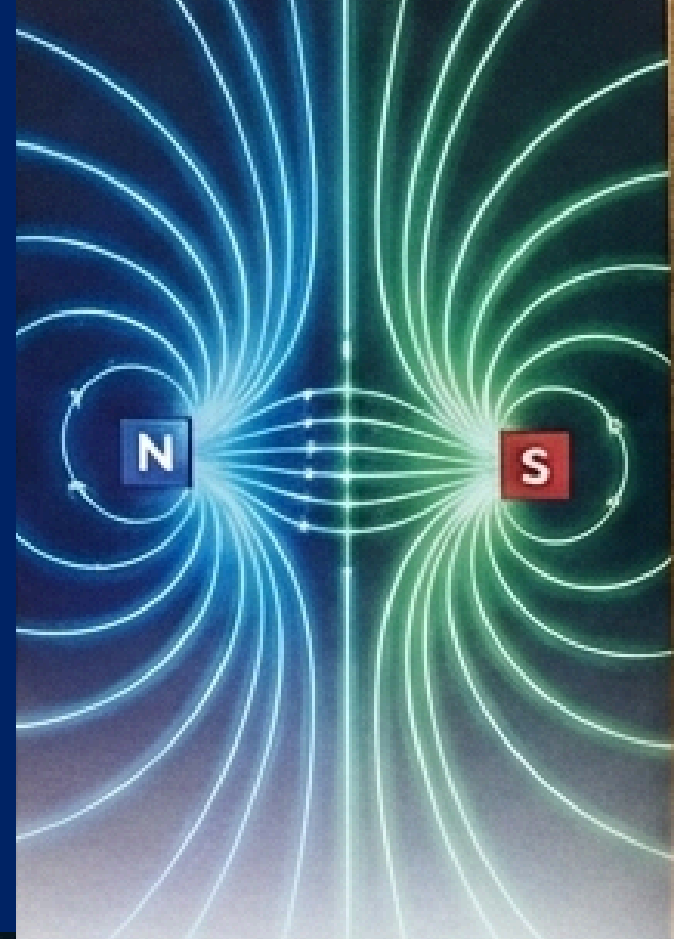
MAGNETIC FIELDS & FORCES



WHAT IS A MAGNETIC FIELD?

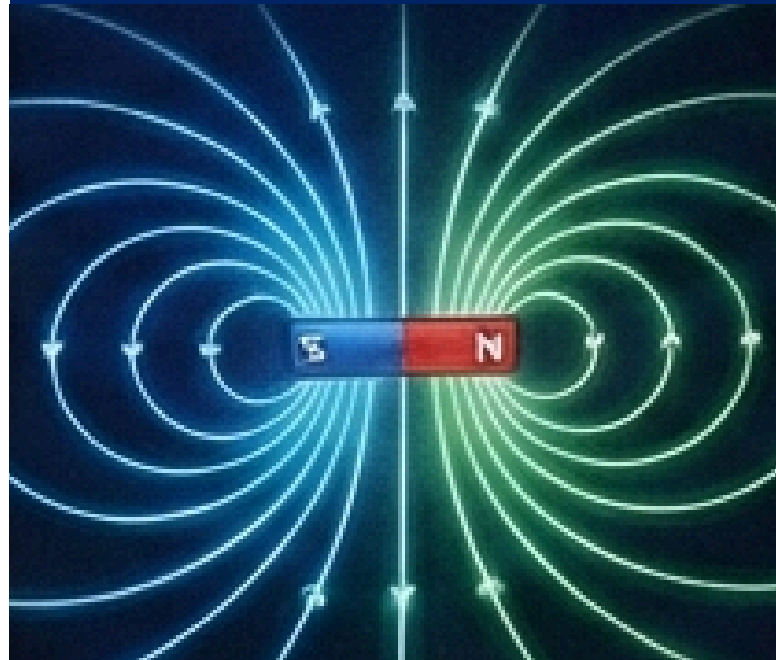


A magnetic field is the invisible area around a magnet where magnetic forces act. It is strongest at the North and South poles and becomes weaker farther away.



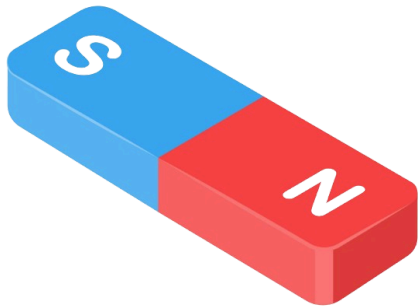
WHAT IS MAGNETIC FORCE?

EMAGNETIC FORCE IS THE PUSH OR PULL THAT MAGNETS CREATE WHEN THEY ATTRACT OR REPEL OTHER MAGNETIC MATERIALS OR MAGNETS.

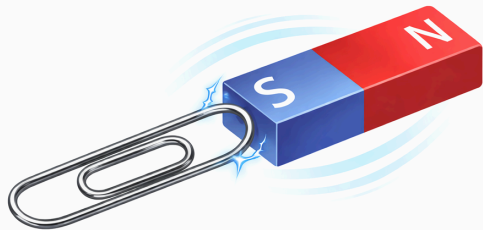


TYPES OF MAGNETS

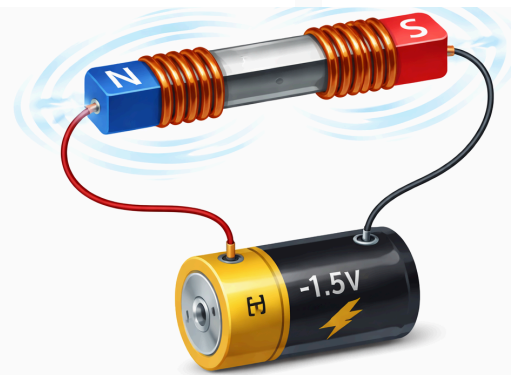
There are three main types of magnets: permanent, temporary, and electromagnets.



PERMANENT



TEMPORARY



ELECTROMAGNETS

Real-Life Applications of Magnetism

- Electric Motors – Used in fans, washing machines, and electric vehicles.
- MRI Machines – Help doctors see inside the human body.
- Compasses – Help with navigation by pointing toward Earth's magnetic north.
- Speakers & Headphones – Convert electrical signals into sound.
- Magnetic Strips – Found on credit cards and ID cards.

FUN FACTS ABOUT MAGNETISM

✨ Earth itself is a giant magnet!

✨ Magnets always have two poles even if you cut them in half.

✨ The strongest natural magnet is called lodestone.

✨ Magnetism is closely related to electricity.

✨ Some animals (like birds and sea turtles) use Earth's magnetic field to navigate!